# TAPE·FOUR

MODEL TF-4 · 4-TRACK CASSETTE MULTITRACKER

# OWNER'S MANUAL

How to record with TAPE·FOUR: the deck, the cassettes, the tape switch, noise reduction, instruments and MIDI. Each control comes with what you should expect to hear when you use it.




FOR WINDOWS · ASIO · VST3

OCTOBER 2026

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# 1 WELCOME

TAPE·FOUR is a four-track cassette recorder that runs on your computer. It records onto a virtual cassette, works with your ASIO audio interface, and plays VST3 instruments as if they were plugged into the deck.

It is deliberately not a DAW. Everything about it follows the workflow of a portable cassette multitracker:

- **Four mono tracks on one side of a tape, all running in one direction.** The tape is a C-46, C-60 or C-90, and the length of your song is limited by the length of the tape.
- **No undo.** Recording replaces whatever was on the tape under the armed tracks. To keep a take, don't record over it.
- **Everything goes to tape as audio.** There are no MIDI tracks. An instrument (a plug-in, one of the built-ins, or a hardware synth) is plugged into a track, you play it, and what you play is recorded as sound.
- **The tape is part of the sound.** Each cassette type has its own headroom, treble limit, hiss and grain, taken from measurements of real tapes. The tape switch, noise reduction, speed and every bounce change the sound, just as they do on a real deck.

#### ABOUT THE SCREENSHOTS

The screenshots in this manual were taken on a test machine with no audio interface or MIDI devices connected. That's why the setup display reads `NO AUDIO DEVICE` and the audio tracks read `INPUT NOT ACTIVE`. On your computer these show your interface and its inputs.

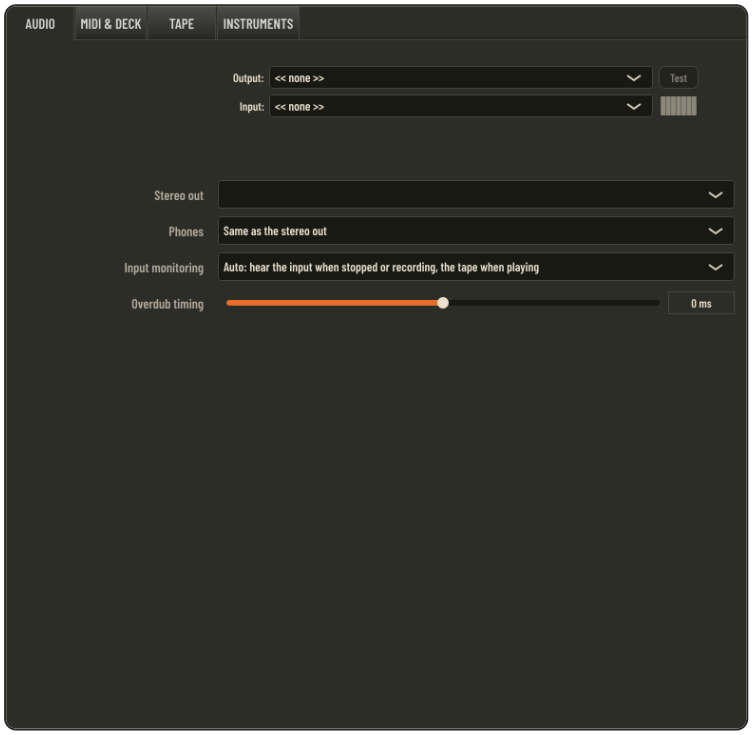
# 2 GETTING STARTED

## INSTALLING

Open the repository's **Actions** tab on GitHub, pick the latest green **Build Windows** run, and download **TapeFour-windows-x64**. Unzip it and run `TapeFour.exe`. Nothing needs installing. Windows may warn that the program comes from an unknown publisher, because it isn't code-signed: choose **More info → Run anyway**.

## AUDIO INTERFACE

Click the amber display at the top right of the deck (`SETUP ›`) to open Setup, then go to the **Audio** tab.




**Setup → Audio.** Choose ASIO and your interface's driver, then the sample rate and buffer size, and the outputs.

| SETTING | WHAT IT DOES | WHAT TO EXPECT |
|---|---|---|
| **Device type, device** | Chooses the driver (ASIO, WASAPI or DirectSound) and the interface. | The setup display on the deck shows the driver, sample rate, buffer size and latency. If an ASIO driver is installed, it is picked automatically the first time you start. |
| **Sample rate, buffer** | The interface's sample rate and block size. | A smaller buffer gives lower latency when you play live, but uses more CPU. 128 samples at 48 kHz is a good starting point. |
| **Stereo out** | The output pair the mix goes to. | The master fader controls this pair. |
| **Phones** | A second output pair, or the same as the stereo out. | The **PHONES** knob on the master strip sets its level, separately from the master fader. |
| **Overdub timing** | ±50 ms trim on top of the latency the driver reports. | Use it if new parts land slightly early or late against what is already on tape. Positive values move recordings earlier. |

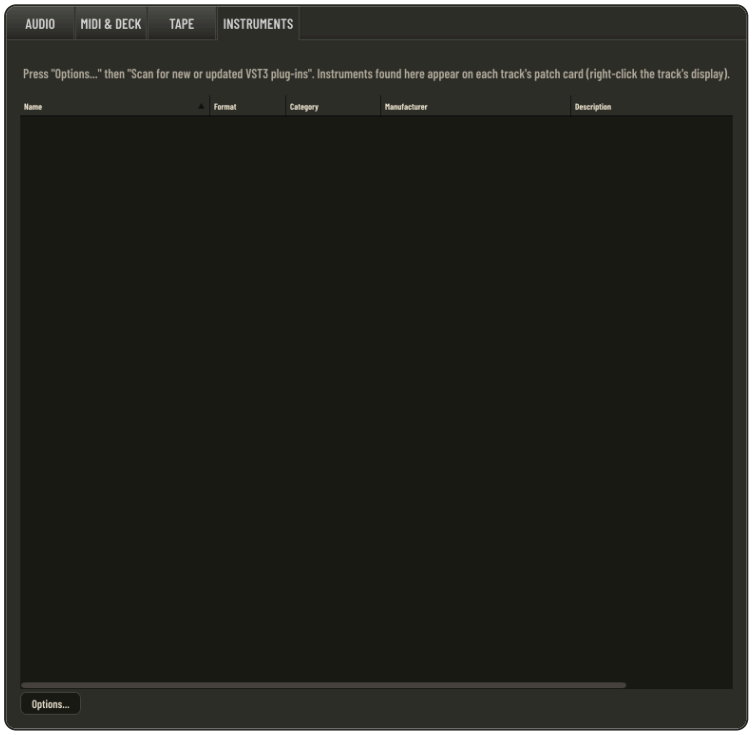
Every change is saved straight away. If the saved driver can't be opened at start-up (for example, because another program is using it), the deck tells you, and keeps your choice for next time.

## MIDI CONTROLLERS AND SYNTHS

In **Setup → MIDI & Deck**, tick the controllers you want to use and the MIDI outputs your hardware synths are on. Chapter 13 covers this in full.

## PLUG-INS

In **Setup → Instruments**, choose **Options… → Scan for new or updated VST3 plug-ins**. Instruments found here appear on each track's patch card. Each plug-in is scanned in a separate helper copy of TAPE·FOUR, so a plug-in that crashes or hangs while being scanned can't take the program down: it is set aside and the scan carries on. TAPE·FOUR also has two built-in instruments, **TF Poly Synth** and **TF Tine Piano**, so you can start without any plug-ins.




**Setup → Instruments.** Scan once; the list is remembered.

# 3 THE FRONT PANEL

The deck is laid out like a portastudio: the mixer runs the full height on the left, and the tape section is a column on the right, with the cassette at the top and the transport below it. The window can be resized; the deck keeps its 16:9 shape.

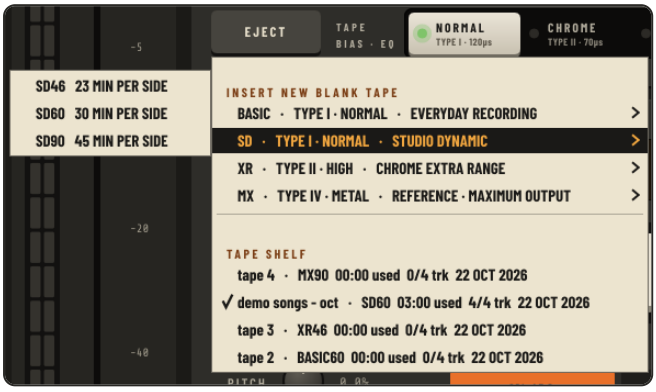



1. **Setup display.** Shows the audio driver and buffer. Click it to open Setup.
2. **MIDI IN / MIDI OUT lamps.** They flash when notes come in from a controller or go out to a hardware synth.
3. **Cassette.** The tape moves from the left spool to the right as it plays. Double-click the label to write on it.
4. **Inlay.** Click it to take the J-card out and write the track list.
5. **EJECT.** Puts in a new blank tape or loads one from the shelf. Stop the tape first.
6. **Tape switch.** NORMAL, CHROME or METAL bias and EQ (chapter 6).
7. **Tape counter**, with **RESET** (set 000 here) and **RTZ** (return to 000).
8. **Status display.** Transport state, time remaining, and warnings such as mismatches.
9. **Transport keys**: REC, REW, F.FWD, PLAY, STOP.
10. **Pitch**: varispeed in steps of 2.4%, up to ±12%.
11. **Tape speed**: 1⅞ or 3¾ inches per second.
12. **Noise reduction** on or off.
13. **NR type display.** Click it to choose the type and the tape character in Setup.
14. **Track display.** What's plugged in, which controller plays it, the audio path, and the NR, EQ and MIDI lamps. Click it to open the track's patch card; double-click it to open its plug-in's window.
15. **Gain, EQ and pan.** HIGH, MID with its FREQ knob, and LOW.
16. **Meter and fader.** 0 VU on the meter is Dolby level on tape.
17. **REC ARM.** An armed track plays its live input; it records when you press RECORD on the transport.
18. **Master strip.** Phones level, stereo meter and master fader.

# 4 CASSETTES

## PUTTING IN A TAPE

With the tape stopped, press **EJECT**. The menu's top half inserts a new blank tape: first choose the cassette (BASIC, SD, XR or MX), then the length. The bottom half, the **tape shelf**, lists every tape you have made. The tape that's in the deck is ticked.




**Inserting a blank.** Choose the cassette, then the length.




**The tape shelf.** Each entry shows the label, cassette, minutes used, tracks used and date.

| LENGTH | PER SIDE | AT 1⅞ IPS | AT 3¾ IPS | SPOOLS |
|---|---|---|---|---|
| **C-46** | 23 min | 23:00 | 11:30 | Smallest packs |
| **C-60** | 30 min | 30:00 | 15:00 | Medium |
| **C-90** | 45 min | 45:00 | 22:30 | Largest packs |

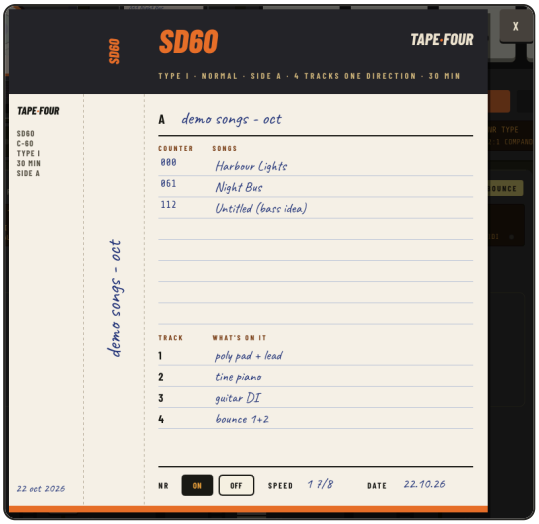
Like a real 4-track, TAPE·FOUR uses one side only, with all four tracks running the same way. When the tape reaches the end, the transport stops and the status display shows `END OF TAPE`.

## SAVING

You never need to save. A tape is written to disk once it has been stopped for a second, and again whenever you change it. It is also saved when you eject it and when you quit. The deck remembers which tape was in when you quit and loads it next time.

## LABEL AND INLAY

Double-click the label on the cassette to write on it. Click the inlay beside the cassette to take the J-card out of its case and write on it: the title, up to eight songs with their counter numbers, a note for each track, whether you recorded with NR, the speed, and the date. Click outside the card, press `Esc` or click **X** to put it back. The title on the inlay and the label on the cassette are the same text, so changing one changes the other.




**The inlay**, opened out: back flap, spine and front. Its colours follow the cassette type.

# 5 THE FOUR TAPE TYPES

Each cassette in TAPE·FOUR behaves like a real tape. Its headroom, treble limit, hiss, grain and output level come from published lab measurements of a real cassette of that kind.

The figures are in dB relative to **Dolby level**, the tape's reference level. On TAPE·FOUR that is **0 VU** on the meters, or −12 dBFS. All figures are at 1⅞ ips without noise reduction.

| TERM | MEANING | WHAT YOU HEAR |
|---|---|---|
| **MOL 400 Hz** | Maximum output level: where a 400 Hz tone reaches 3% distortion. | How loud you can record before the tape thickens and then distorts. Bass gets a little less room than the mids, as on real tape. |
| **SOL 10 kHz** | Saturation output level for treble. | How loud cymbals, hi-hats and bright synths can go before the tape squashes them. Treble saturates long before the mids, especially at 1⅞ ips. |
| **S/N** | Hiss on blank tape, A-weighted, below Dolby level. | The hiss in quiet passages. |
| **Mod. noise** | Modulation noise: grain that rides on the signal. | A fine roughness on sustained notes. It gets louder and quieter with the music, so noise reduction can't remove it. |
| **Sensitivity** | How loud the tape plays back what you recorded. | A small level shift on every pass through the tape. You notice it after several bounces. |
| **EQ** | The playback EQ time constant: 120 µs or 70 µs. | 70 µs tapes have less treble hiss. |




## BASIC

TYPE I · NORMAL BIAS · 120 µs · MODELLED ON SONY HF

A budget ferric tape, the kind that came in multipacks. It has the least headroom and the most hiss and grain of the four. It sounds warm and slightly fuzzy, and compresses early if you record hot. It's good for sketches, lo-fi textures and anything you want to sound like an old demo.

| | |
|---|---|
| MOL 400 / 100 Hz | +2.4 / +2.0 dB |
| SOL 10 kHz | −10.3 dB |
| S/N | 54.3 dB(A) |
| Mod. noise | −36.1 dB |
| Sensitivity | −1.0 dB |




## SD

**TYPE I · NORMAL BIAS · 120 µs · MODELLED ON TDK AD**

A premium ferric tape and the all-rounder. It has good mid-range headroom, the best treble headroom of the Type I tapes, and moderate hiss. Tapes made before tape types existed play as SD. It's the tape to reach for when you don't have a reason to pick another.

| | |
|---|---|
| MOL 400 / 100 Hz | +6.2 / +3.8 dB |
| SOL 10 kHz | −8.2 dB |
| S/N | 60.3 dB(A) |
| Mod. noise | −44.3 dB |
| Sensitivity | −0.3 dB |




## XR

**TYPE II · HIGH (CHROME) BIAS · 70 µs · MODELLED ON TDK SA**

A chrome-position tape. Its 70 µs playback EQ makes the hiss quieter and less bright, and it plays back slightly hot. It has the least treble headroom of the four, so loud bright sounds dull and compress first, especially at 1⅞ ips. It's clean and quiet for vocals, acoustic instruments and pads. For bright, loud material, use 3¾ ips or MX.

| | |
|---|---|
| MOL 400 / 100 Hz | +4.4 / +3.4 dB |
| SOL 10 kHz | −11.2 dB |
| S/N | 60.9 dB(A) |
| Mod. noise | −47.2 dB |
| Sensitivity | +1.1 dB |




## MX

**TYPE IV · METAL BIAS · 70 µs · MODELLED ON MAXELL MX**

A metal tape. It has the most headroom everywhere (about 3 dB more mid-range and 1.4 dB more treble than SD), the lowest grain and the lowest hiss. It takes hot levels and bright material without complaint. It's the closest a cassette gets to transparent.

| | |
|---|---|
| MOL 400 / 100 Hz | +9.1 / +8.0 dB |
| SOL 10 kHz | −6.8 dB |
| S/N | 62.7 dB(A) |
| Mod. noise | −50.4 dB |
| Sensitivity | +0.8 dB |

## CHOOSING A TAPE: WHAT TO EXPECT

| IF YOU… | BASIC | SD | XR | MX |
|---|---|---|---|---|
| **record a vocal peaking at +3 VU** | Audibly thick, a little gritty | Clean, slightly rounded | Clean | Clean |
| **record hi-hats at 0 VU at 1⅞** | Softened, a bit dull | Mostly intact | Noticeably dulled | Intact |

| IF YOU… | BASIC | SD | XR | MX |
|---|---|---|---|---|
| **listen to blank tape (NR off)** | Most hiss | About 6 dB less | Less, and darker | Least |
| **hold a long synth note** | Slight roughness | Faint | Very faint | Practically none |
| **bounce three times** | About 3 dB quieter | About 1 dB quieter | About 3 dB louder | About 2.4 dB louder |

Generation loss comes from each tape's sensitivity, which is applied on every pass. Noise reduction doubles it (chapter 11).

# 6 THE TAPE SWITCH

The **TAPE · BIAS · EQ** switch under the cassette sets three things at once, as on a real deck: the record bias, the record EQ and the playback EQ.




**A mismatch.** An MX (metal) tape is loaded but the switch is at NORMAL. The lamp on NORMAL is amber, and the status display reads `BIAS LOW: TYPE IV TAPE AT NORMAL`.

- **When a tape goes in, the switch moves to that tape's type.** After that it stays where you put it, so a mismatch is always your choice. Restarting the audio device doesn't move it.
- **One lamp shows where the switch is.** It is green when that position suits the loaded cassette and amber when it doesn't. The status display then says which way the bias is off.
- **Every stretch of tape remembers the switch position it was recorded at**, just as it remembers noise reduction and speed.

## BIAS: RECORDING AT THE WRONG POSITION

A Type II tape needs about 1.5 times the bias of a Type I, and a Type IV about 2.5 times.

- **Too little bias (under-bias)** leaves the tape bright and fuzzy. It distorts and compresses sooner and gets grainier.
- **Too much bias (over-bias)** wipes out the treble as it records. The tape sounds dull and a little quieter, but slightly cleaner.

These effects are recorded into the tape. The table assumes you play the tape back at the same position you recorded at.

| TAPE ↓ · SWITCH → | NORMAL | CHROME | METAL |
|---|---|---|---|
| **Type I: BASIC, SD** | Correct. | Over-biased by 3.5 dB. About −7 dB at 10 kHz, −0.9 dB overall, slightly cleaner (+0.9 dB MOL). Muffled. | Over-biased by 8 dB. About −16 dB at 10 kHz and −2 dB overall. Very dull, like a worn head. |
| **Type II: XR** | Under-biased by 3.5 dB. About +2.6 dB at 10 kHz, MOL −5.3 dB, grain +4 dB. Bright and edgy; distorts early. | Correct. | Over-biased by 4.5 dB. About −9 dB at 10 kHz and −1.1 dB overall. Dull. |
| **Type IV: MX** | Under-biased by 8 dB. MOL −12 dB, grain +10 dB, −2.5 dB overall, treble about −1 dB. Heavily distorted and gritty. **Old recordings bleed through** (see below). | Under-biased by 4.5 dB. About +2.5 dB at 10 kHz, MOL −6.8 dB, grain +5 dB. Bright and crunchy. | Correct. |

**ABOUT THESE FIGURES**

The bias ratios between tape types follow the tape standards. How much the treble, headroom and grain move per dB of bias error is an estimate of typical real-deck behaviour, not a published measurement. The automated tests check every row of this table.

## PLAYBACK EQ: PLAYING BACK AT THE OTHER POSITION

NORMAL plays back with 120 µs EQ, while CHROME and METAL both use 70 µs. Each stretch of tape is played back through the EQ of the switch as it is now, compared with the EQ it was recorded with:

| RECORDED AT | PLAYED AT | RESULT |
|---|---|---|
| **CHROME or METAL (70 µs)** | NORMAL (120 µs) | Treble lifted by a shelf starting near 1.3 kHz: about +4.5 dB at 8 kHz, up to +4.7 dB. Bright and hissy. |
| **NORMAL (120 µs)** | CHROME or METAL (70 µs) | The reverse: about −4.5 dB at 8 kHz. Muffled. |
| **CHROME** | METAL, or the other way round | No EQ change; both use 70 µs. |

Blank-tape hiss follows the same rule: a Type II or IV tape played at NORMAL hisses brighter. You can combine both effects. For example, record a chrome tape at NORMAL (under-biased and bright), then play it back at CHROME.




**The EQ lamp.** These tracks were recorded at NORMAL and the switch is now at CHROME, so each track's EQ lamp is red and the switch lamp is amber. It flashes while the tape plays. It is green when the recording's EQ matches the switch, and off on blank tape.

## METAL TAPE AT NORMAL: OLD MATERIAL BLEEDS THROUGH

On a NORMAL setting the erase head isn't strong enough to fully erase high-coercivity metal tape. If you record over an MX tape at NORMAL, about 30 dB of the old recording survives underneath the new one, mostly its bass. Treble erases more easily. Record over it at METAL and the old recording is fully gone (it measures below −90 dB).

**TRY THIS**

Record a bass line on an MX tape at METAL. Set the switch to NORMAL and record a new part over it on the same track. Play it back: the old bass line is still there underneath, faint and muffled, like a ghost of the earlier take.

# 7 TRANSPORT AND COUNTER

| CONTROL | WHAT IT DOES | WHAT TO EXPECT |
|---|---|---|
| **PLAY** `Space` | Plays the tape. | Unarmed tracks play the tape; armed tracks keep playing their live input, so you can rehearse over the tape. The PLAY lamp lights. |
| **STOP** `Space` | Stops. | One second later the tape is saved. |
| **REC** `R` | Records on every armed track. | The REC lamp lights. If no track is armed, the status display says `ARM A TRACK FIRST`. |
| **REC while playing** | Punch-in. | Recording starts at once on the armed tracks. Press PLAY to punch out and keep playing. Tape before and after the punch is untouched. |
| **REW, F.FWD** | Wind the tape. | The tape speeds up and slows down like real spools. Winding stops at either end of the tape. |
| **RESET** | Sets the counter to 000 where the tape is now. | Use it to mark the start of a song. |
| **RTZ** `Home` | Return to zero. | Rewinds and stops at 000. The status display reads `RETURN TO ZERO`. |

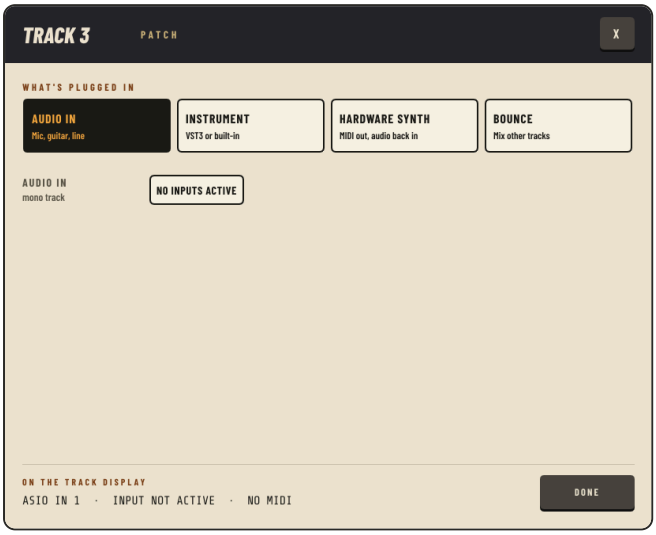
The counter is a three-digit mechanical-style counter. Like the counters on real decks, it doesn't count seconds: it advances one count for every two seconds of tape at 1⅞ ips. Write the counter number for each song on the inlay. The status display shows the time remaining at the current speed.

**NO UNDO**

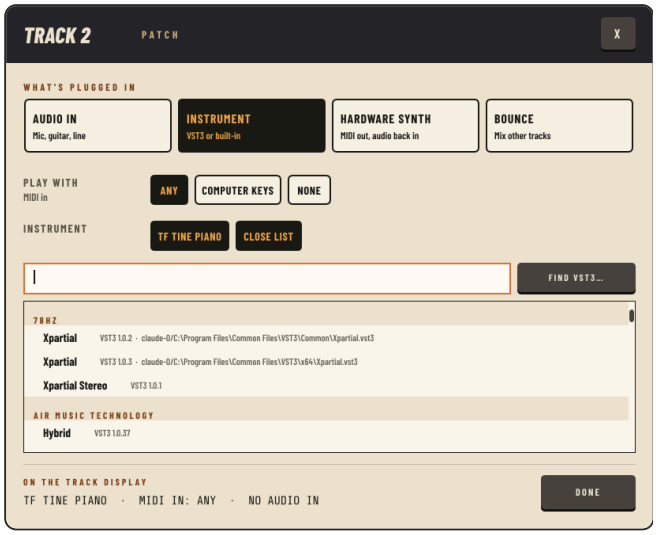
Recording replaces what was on the tape under the armed tracks, from the moment you press REC until you stop or punch out. Nothing else is touched. To keep a take, record on a different track or a new tape.

# 8 PATCHING A TRACK

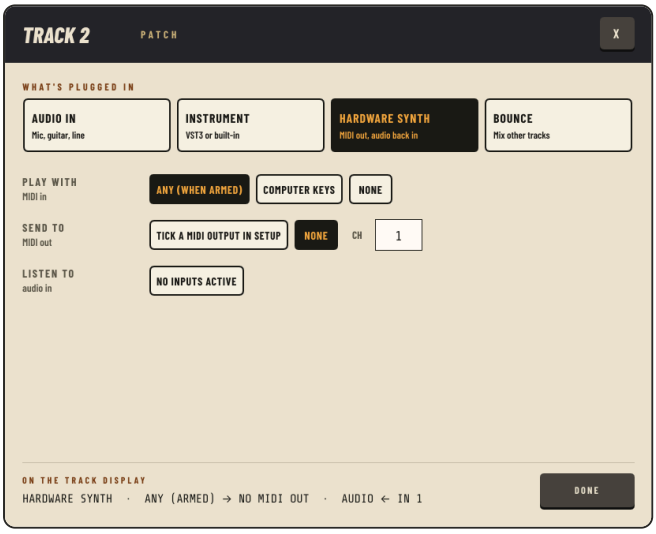
Click or right-click a track's display to open the track's **patch card**. The card shows what's plugged into the track, what plays it, and where its audio comes from. At the bottom, it previews what the track display will say. Click **DONE**, click outside the card or press `Esc` to close it. A track has exactly one source: plugging in a new one unplugs the old.



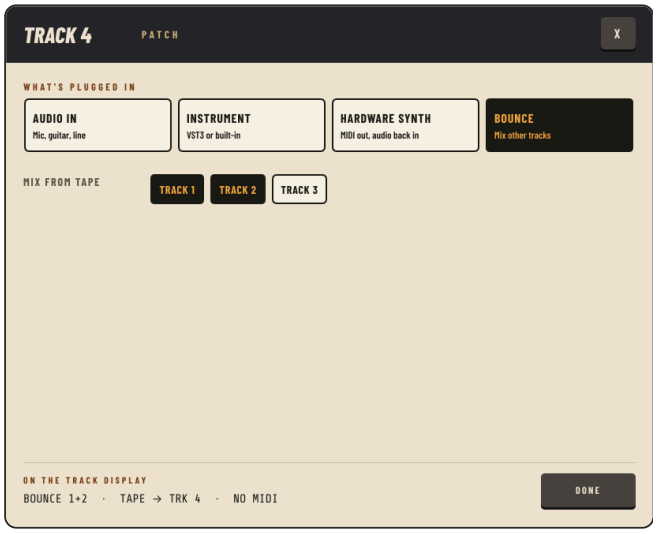

**Audio in.** Choose an input, or a pair of inputs mixed to mono.




**Instrument.** CHANGE... opens a searchable list of every instrument; EDIT SOUND opens a plug-in's window.




**Hardware synth.** MIDI goes out to the synth and its audio comes back in on an input.




**Bounce.** Mix other tracks from tape onto this one.

| SOURCE | SETTINGS | WHAT TO EXPECT |
|---|---|---|
| Audio in | **AUDIO IN**: **IN N**, or **IN N+M MONO**. | The display shows `ASIO IN 1` and the input's name. Tracks are mono; a stereo source on a pair is summed to mono at half level per side. |

| SOURCE | SETTINGS | WHAT TO EXPECT |
|---|---|---|
| **Instrument** | **PLAY WITH**: **ANY**, a named controller, or **NONE**.<br>**INSTRUMENT**: TF Poly Synth, TF Tine Piano, any scanned VST3, chosen from the list (**CHANGE…**),<br>**EDIT SOUND**. | An instrument only sounds while its track is armed. **Any**: any ticked controller plays it. **A named controller** plays only that track, so different keyboards can play different armed tracks at once. **CHANGE…** opens a searchable list of the built-in instruments and every scanned VST3, grouped by maker; type to filter and press `Return` for the first match. Two plug-ins with the same name are both listed, with their folder and file so you can tell them apart. **EDIT SOUND** (or double-clicking the track display) opens the plug-in's own window to change its sound; the window stays on top, and `Space`, `R` and `Home` still work the transport while it has focus. Plug-ins come in 12 dB below their own output, so a full-level synth sits near 0 VU. |
| **Hardware synth** | **PLAY WITH** (as above)<br>**SEND TO**: a MIDI output, and a channel 1–16, or 0 for "as played".<br>**LISTEN TO**: the input the synth is plugged into. | Notes go out to the synth on the chosen channel, and its audio is recorded from the input. An MPE controller's notes pass through on their own channels. Changing the route sends all-notes-off so nothing hangs. A track can't play a hardware synth and a plug-in at once. |
| **Bounce** | **MIX FROM TAPE**: tick the source tracks. | While recording, you hear the sources once, through their own EQ, pan and fader. Use those to balance the bounce, then arm the destination and record. The bounce picks up another pass of the tape: a little more hiss and grain, and a level shift of the tape's sensitivity. |

## READING THE TRACK DISPLAY




**Track displays.** Line 1: the source. Line 2: what plays it, or the input's name. Line 3: the audio path, plus three lamps.

| LAMP | LIT | RED |
|---|---|---|
| **NR** | The tape under the head was recorded with noise reduction, or you're recording with NR on. | Flashes red while playing if the NR switch doesn't match how the tape was recorded. |
| **EQ** | Green: the tape here was recorded with the same playback EQ as the switch. | The switch's EQ differs from the recording's (chapter 6). It flashes while playing. |
| **MIDI** | Green flash: notes are reaching this track. | — |

When the tape under the head was recorded at a different speed from the one you're playing at, line 3 shows the pitch shift in semitones, for example `-12.0 SEMITONES`.

# 9 THE MIXER

Each of the four channel strips has its controls in a single column, top to bottom. Double-click any knob or fader to reset it: gain to 0 dB, EQ and pan to centre, FREQ to 1.2 kHz, fader to the 0 mark.

| CONTROL | RANGE | WHAT TO EXPECT |
|---|---|---|
| **GAIN** | −20 to +40 dB (0 dB at the default) | Input trim before the tape. Use it to put peaks near 0 VU. Higher levels reach the tape's MOL and compress or distort (chapter 5). |
| **HIGH** | ±12 dB shelf: about ±11.6 dB at 10 kHz, ±7 dB at 5 kHz, half way at 4.5 kHz | Presence and air, well inside what the tape can record. Full boost is clearly brighter, and also brings up hiss. |
| **MID** | ±12 dB peak | Presence, honk or boxiness, depending on FREQ. A cut makes a part sit back. |
| **FREQ** | 250 Hz to 5 kHz (1.2 kHz at the centre) | Where MID works; the smaller knob with the blue pointer. Around 250–500 Hz for body or mud, 1–2 kHz for honk, 3–5 kHz for bite and vocal presence. Sweep it with MID boosted to find a spot, then cut or boost. |
| **LOW** | ±12 dB shelf: about ±11.6 dB at 100 Hz, ±6 dB at 250 Hz | Weight. A cut helps when bouncing several tracks together. |
| **PAN** | L100 to R100, constant power | Centre is −3 dB to each side. Hard left puts the full level on the left output only. |
| **Fader** | −∞ to +7 dB (0 at the mark) | The level into the mix and into bounces. On an armed track it also sets the level going to tape, as on a portastudio. |
| **Meter** | 20 LEDs, 0 VU at −12 dBFS | Green below 0 VU, amber from 0 to +3, red above +3. Keep peaks around 0 VU for a clean recording, and push into the amber for tape colour. |
| **REC ARM** | On or off | An armed track plays its live input (audio in, instrument or hardware synth) instead of its tape, whatever the transport is doing, and the display reads `ARMED: LIVE INPUT`. Press RECORD on the transport and it records that input to tape. To hear a take, disarm the track: while it is armed you hear the input, not the tape. |

The **master strip** has a stereo meter, the master fader for the stereo out, and the **PHONES** knob, which sets the level of the phones output pair independently.

## WHAT THE EQ AND GAIN RECORD, AND WHAT THEY DON'T

On a cassette 4-track the channel strip sits between the input and the tape. TAPE·FOUR works the same way: gain, EQ and the fader shape what is recorded on an armed track. On playback, the same strip shapes what you hear from the tape. To fix a recording later, adjust the strip on playback, or bounce it through the strip onto another track.

# 10 SPEED AND PITCH

| SETTING | WHAT IT DOES | WHAT TO EXPECT |
|---|---|---|
| **1⅞ ips** | Normal cassette speed. | The tape's full length. Treble saturates sooner and there's slightly more hiss. |
| **3¾ ips** | Double speed, as on many 4-track portastudios. | Half the recording time. About an octave more treble headroom (the treble limit moves up), and about 3 dB less hiss in the audible range. |
| **Playing at the other speed** | Every stretch of tape remembers the speed it was recorded at. | A recording made at 3¾ and played at 1⅞ sounds an octave lower at half speed, and the reverse sounds an octave higher. The track display shows the shift in semitones. Record at 1⅞ and play at 3¾ for chipmunk vocals; do the reverse for slow, dark textures. |
| **PITCH** | Varispeed, ±5 steps of 2.4% (up to ±12%). | Speeds up or slows down the tape and changes pitch and tempo together, by about 0.4 semitone per step. Use it to tune the tape to an instrument, or to record a part slowed down and play it back at normal speed. |

Wow, flutter and drift (chapter 12) are added on top of the pitch setting, so the tape always moves slightly unevenly unless you turn them off.

# 11 NOISE REDUCTION

Noise reduction encodes what you record and decodes on playback. On the way to tape it boosts quiet sounds, mostly the treble. On the way back it turns them down again, and the tape hiss goes down with them. Turn it on with the **NOISE REDUCTION** button. Choose the type by clicking the **NR TYPE** display, or in **Setup → Tape**.

| TYPE | HOW IT WORKS | HISS REDUCTION | CHARACTER |
|---|---|---|---|
| **2:1 compander** | Broadband 2:1 compression on record and 1:2 expansion on playback, with fixed treble pre-emphasis (dbx-style). | The most, roughly 30 dB | Very quiet. Can "pump": hiss and grain rise and fall behind loud bass notes. |
| **Sliding band B** | Lifts quiet treble up to 10 dB above about 1.5 kHz. | About 10 dB | Gentle and forgiving. |
| **Sliding band C** | A deeper band from about 700 Hz, up to 20 dB. | About 20 dB | Quieter than B, and more sensitive to level and treble errors. |

## MISMATCHED NR

Tape remembers whether it was recorded with NR. If you play it back with the switch the other way, the track's NR lamp flashes red. Encoded tape played without NR sounds bright and squashed, and the treble pumps. Plain tape played with NR sounds dull and lifeless.

## HOW THE TAPE TYPE AND NOISE REDUCTION INTERACT

Noise reduction expects the tape to give back exactly what the encoder put on it. Anything that changes level or treble on the way through gets enlarged by the decoder, not just passed along. In TAPE·FOUR the encoder sits before the tape model and the decoder after it, so all of this happens on its own:

- **Hiss floor.** NR removes a fixed amount, so the tape's own hiss still matters. MX with C or 2:1 is the quietest combination, and BASIC with B the noisiest.
- **Level errors grow.** With 2:1, a 1 dB difference becomes 2 dB. SD (−0.3 dB) loses about 0.6 dB per pass with NR on. XR (+1.1 dB) gains about 2.2 dB. B and C mistrack in the treble on quiet passages.
- **Treble errors grow and move.** A bias or EQ mismatch is a treble error. B and C read it as a level change, so a bright mismatch gets brighter on quiet passages and a dull one duller. The treble "breathes" with the music. With 2:1 the treble error is doubled throughout.
- **Treble saturation.** The encoder lifts quiet treble before the tape. On tapes with little treble headroom (XR and BASIC, especially at 1⅞ ips), loud bright parts saturate after encoding. The decoder then mistracks, and cymbals turn dull and swishy. MX has the most room.
- **Grain.** NR can't remove modulation noise, which rides on the signal. A 2:1 compander's gain swings make it, and the hiss, rise and fall. That's most obvious on BASIC.

# 12 TAPE CHARACTER

**Setup → Tape** shows the loaded cassette's measured figures. Its sliders adjust the deck and the tape's condition on top of those figures. Double-click **HISS** or **TAPE DRIVE** to return it to the centre.




**Setup → Tape**, with an SD60 loaded.

| CONTROL | RANGE | WHAT TO EXPECT |
|---|---|---|
| **Hiss** | Off, or −12 to +12 dB relative to the tape's measured hiss (centre = as measured) | Real cassettes without NR hiss at 54–63 dB below 0 VU. That is audible in quiet passages but not intrusive. Turn it up to taste for a dirtier tape, or off for a silent background. |
| **Wow** | 0–0.5% | A slow pitch waver (under 1.5 Hz). It's most audible on sustained notes and piano. |
| **Flutter** | 0–0.3% | A fast waver (6–12 Hz). It gives a slight shimmer, and above 0.15% a watery roughness. |
| **Tape drive** | −6 to +6 dB (centre = 0 VU at Dolby level) | Where 0 VU lands on the tape. More drive brings saturation and treble compression earlier, the way a deck calibrated hot would. Less drive keeps it cleaner. The playback level isn't changed. |
| **HF loss** | 0 to −12 dB at 10 kHz | A worn or misaligned head. It dulls everything played back, hiss included. |
| **Track crosstalk** | Off, or −80 to −30 dB | Neighbouring tracks leak into each other. You may hear it as a faint ghost of track 2 behind track 1. |
| **Dropouts** | None to frequent | Brief dips in level, from 15 to 75 ms, as a damaged patch of tape passes the head. |
| **Speed drift** | 0 to ±1% | The tape speed wanders slowly, every few seconds, as with tired motor belts. |

| CONTROL | RANGE | WHAT TO EXPECT |
|---|---|---|
| **Noise reduction** | 2:1, B or C | See chapter 11. |

## CONDITION PRESETS

The presets set every slider except Tape drive, which is calibration rather than wear.

| PRESET | HISS | WOW | FLUTTER | HF LOSS | CROSSTALK | DROPOUTS | DRIFT |
|---|---|---|---|---|---|---|---|
| **Clean** | off | 0 | 0 | 0 dB | off | none | 0 |
| **Well kept** | +1.2 dB | 0.15% | 0.08% | −2.4 dB | −65 dB | rare | ±0.2% |
| **Worn** | +6 dB | 0.30% | 0.15% | −6.6 dB | −55 dB | occasional | ±0.4% |
| **Chewed** | +9.6 dB | 0.45% | 0.24% | −9.6 dB | −45 dB | frequent | ±0.7% |

# 13 MIDI AND THE DECK STATE




**Setup → MIDI & Deck.** Controllers in, hardware synths out, and deck recall.

## CONTROLLERS IN

- **Use**: TAPE·FOUR only listens to controllers that are ticked. Each track then chooses its controller on its patch card.
- **MPE**: tick it for MPE controllers (Osmose, Seaboard, LinnStrument and others). Known MPE controllers start ticked. A track played by an MPE controller tells its instrument about the MPE zone (lower zone, 15 channels). Each note then gets its own pitch bend, pressure and timbre (CC74). The built-in instruments respond directly. VST3s need MPE support, and some need it switched on in their own settings.
- **Bend range**: type the per-note pitch-bend range your controller sends (the Osmose default is 48 semitones). If it's wrong, bent notes land at the wrong pitch.
- **Computer keyboard**: `A W S E D F T G Y H U J K O L P ;` play notes like a piano keyboard, and `Z` and `X` change octave.

## PLUGGING IN WHILE RUNNING

TAPE·FOUR notices MIDI devices being plugged in and out while it runs. Controllers and synth outputs you have ticked reconnect on their own, and tracks routed to them carry on. A controller TAPE·FOUR has never seen comes in switched on; untick it in Setup if you don't want it.

## HARDWARE SYNTHS OUT

Tick the MIDI outputs your synths are on. You can give each one a name, such as "JUNO-106", which the track display then shows instead of the port name. The MIDI OUT lamp at the top of the deck flashes when notes go out.

## RECALL THE DECK WITH EACH TAPE

Every tape remembers the deck it was last used with:

- sources, and instruments with their plug-in settings;
- MIDI routing;
- gain, EQ, pan and faders;
- master and phones;
- speed and pitch;
- tape character.

With **Recall the deck with each tape** ticked, loading a tape puts the deck back exactly as it was. Unticked (the default), the deck stays as it is when you change tapes. The tape switch always moves to the new tape's type either way.

# 14 REFERENCE

## KEYBOARD

| KEY | ACTION |
|---|---|
| `Space` | Play / stop |
| `R` | Record (punch in while playing) |
| `Home` | Return to zero |
| `Esc` | Close the patch card or inlay |
| `A` ... `;` | Play notes; `Z` / `X` change octave |

## WHERE THINGS ARE KEPT

| | |
|---|---|
| **Tapes** | `Documents\TapeFour\Tape Shelf\`: one folder per tape, holding the audio for each track and `tape.xml` (label, inlay, flags and deck state). |
| **Settings** | `%APPDATA%\TapeFour\TapeFour.settings`: audio device, MIDI, deck and tape character. |

## TROUBLESHOOTING

| PROBLEM | TRY THIS |
|---|---|
| `NO AUDIO DEVICE` | Open Setup → Audio and choose your driver. If another program holds the ASIO driver, close it, then reselect the driver. |
| **REC does nothing** | Arm at least one track. The status display says `ARM A TRACK FIRST`. |
| **A take seems to have recorded silence** | While a track is armed you hear its live input, not the tape. Disarm it and play the take back. |
| **An instrument doesn't play** | Tick the controller in Setup → MIDI & Deck. Check the track's **PLAY WITH** setting, and arm the track: instruments only sound while armed. Watch the MIDI IN lamp and the track's MIDI lamp. |
| **Recordings sound distorted** | Lower GAIN so peaks sit around 0 VU, use a tape with more headroom (MX), centre Tape drive, and check the tape switch matches the tape. |
| **Everything sounds dull** | Check the EQ lamps (playback EQ mismatch), the NR lamps (NR mismatch), the tape switch (over-bias) and HF loss in Setup → Tape. |
| **Bounces get quieter or louder** | That's the tape's sensitivity, doubled with 2:1 NR. Correct it with the faders before each bounce. |
| **A plug-in shows as failed after a scan** | It crashed or hung while being scanned, so TAPE·FOUR set it aside. Update the plug-in, select it in Setup → Instruments, choose **Options... → Remove selected plug-in from list**, and scan again. |
| **A controller plugged in later doesn't play** | Check it's ticked in Setup → MIDI & Deck, and that the track's **PLAY WITH** names it or is set to **Any**, and that the track is armed. |

| PROBLEM | TRY THIS |
|---|---|
| **A plug-in is missing when a tape is recalled** | Rescan in Setup → Instruments. The plug-in's settings are kept until it can be loaded again. |
| **A ghost of an old take is audible** | You recorded over a metal tape at NORMAL. Record over it again at METAL to erase it fully. |

## CASSETTE MEASUREMENTS

| CASSETTE | TYPE | MEASURED ON | MOL 400 | MOL 100 | SOL 10K | S/N (A) | MOD. NOISE | SENS. | EQ |
|---|---|---|---|---|---|---|---|---|---|
| **BASIC** | I | Sony HF | +2.4 | +2.0 | −10.3 | 54.3 | −36.1 | −1.0 | 120 µs |
| **SD** | I | TDK AD | +6.2 | +3.8 | −8.2 | 60.3 | −44.3 | −0.3 | 120 µs |
| **XR** | II | TDK SA | +4.4 | +3.4 | −11.2 | 60.9 | −47.2 | +1.1 | 70 µs |
| **MX** | IV | Maxell MX | +9.1 | +8.0 | −6.8 | 62.7 | −50.4 | +0.8 | 70 µs |

dB relative to 400 Hz Dolby level (0 VU), at 1⅞ ips, without noise reduction. TAPE·FOUR's automated tests record tones through each cassette and check that these figures come out within about 1 dB, and that the tape switch's EQ and bias effects match chapter 6.

## SOURCES


- Compact Cassette tape types and formulations, Wikipedia (including Roberson's measurements of retail cassettes): en.wikipedia.org/wiki/Compact_Cassette_tape_types_and_formulations
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